

Boundary Layer Meteorology Reference: Stability Functions Based Upon Shear Functions

Equation Reference Guide (Adapted from England & McNider, 1995)

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1 Exchange Coefficients & Surface Layer Framework

The Monin–Obukhov exchange coefficients for momentum (K_m) and heat (K_h) are defined as:

$$K_m = \frac{l_m u_*}{\phi_m} = \frac{l_m^2 s}{\phi_m^2} \quad (1)$$

$$K_h = \frac{l_m u_*}{\phi_h} = \frac{l_m^2 s}{\phi_m \phi_h} \quad (2)$$

where friction velocity u_* , wind shear s , mixing length l_m , and temperature scale θ_* are:

$$u_* = \frac{l_m s}{\phi_m}, \quad s = \left[\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 \right]^{1/2} \quad (3)$$

$$l_m = \kappa z, \quad \theta_* = \frac{l_m}{\phi_h} \frac{\partial \theta}{\partial z} \quad (4)$$

The dimensionless shear functions $\phi(\zeta)$ depend on stability parameter $\zeta = z/L$:

- **Stable branch** ($\zeta > 0$):

$$\phi_m(\zeta) = 1 + \beta_m \zeta, \quad \phi_h(\zeta) = 1 + \beta_h \zeta \quad (5)$$

- **Unstable branch** ($\zeta < 0$):

$$\phi_m(\zeta) = (1 - b_m \zeta)^{-1/4}, \quad \phi_h(\zeta) = (1 - b_h \zeta)^{-1/2} \quad (6)$$

Notation Note: Empirical constants β_m, β_h (stable) and b_m, b_h (unstable) are distinct experimental parameters. Stable parameters are denoted β throughout for consistency with England & McNider (1995).

2 Gradient Richardson Number (Ri) & Inverse Relations

The gradient Richardson number is related to Monin–Obukhov stability ζ via:

$$Ri = \frac{g}{\theta_0} \frac{\partial \theta / \partial z}{s^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \quad (7)$$

Stable Boundary Layer ($\zeta > 0$)

Substituting stable shear functions into $Ri(\zeta)$ yields the quadratic equation:

$$(Ri\beta_m^2 - \beta_h)\zeta^2 + (2Ri\beta_m - 1)\zeta + Ri = 0 \quad (8)$$

The physically continuous solution requiring $\zeta \rightarrow 0$ as $Ri \rightarrow 0$ (neutral limit) is:

$$\zeta = \frac{1 - 2Ri\beta_m - \sqrt{1 + 4(\beta_h - \beta_m)Ri}}{2(Ri\beta_m^2 - \beta_h)} \quad (9)$$

- **MOST Singular Richardson Number:**

$$Ri_c^{\text{MOST}} = \frac{\beta_h}{\beta_m^2} \quad (10)$$

This value corresponds strictly to the singular limit of the denominator in the stable MOST inverse relation ($\zeta \rightarrow +\infty$). It should not, in general, be interpreted as a dynamically universal critical Richardson number or as the point where physical turbulence collapses ($Ri_c^{\text{MOST}} \neq Ri_{\text{fold}}^{\text{GSPT}}$).

- **Discriminant vs. Denominator Singularities:** For general $\beta_h \neq \beta_m$, the inverse root exhibits two distinct algebraic properties:

1. Denominator vanishing at $Ri = \beta_h/\beta_m^2$, sending $\zeta \rightarrow +\infty$.
2. Discriminant vanishing when $\Delta = 1 + 4(\beta_h - \beta_m)Ri = 0$, occurring at $Ri = -1/[4(\beta_h - \beta_m)]$ if $\beta_h < \beta_m$.

- **Exact Linear Simplification ($\beta_h = \beta_m = \beta$):** When heat and momentum constants match, the discriminant collapses to $\Delta \equiv 1$ for all Ri , removing the radical completely:

$$\zeta = \frac{Ri}{1 - Ri/Ri_c^{\text{MOST}}} \quad \text{where } Ri_c^{\text{MOST}} = \frac{1}{\beta} \quad (11)$$

Unstable Boundary Layer ($\zeta < 0$)

Squaring and clearing denominators for the unstable branch yields the cubic equation:

$$b_m\zeta^3 - \zeta^2 - b_h Ri^2\zeta + Ri^2 = 0 \quad (12)$$

3 Stability Functions $f_m(Ri)$ and $f_h(Ri)$

In numerical models, exchange coefficients are expressed directly in terms of Ri :

$$K_m = f_m(Ri)l_m^2 s, \quad K_h = f_h(Ri)l_m^2 s \quad (13)$$

where:

$$f_m(Ri) = \phi_m^{-2}(\zeta), \quad f_h(Ri) = \phi_m^{-1}(\zeta)\phi_h^{-1}(\zeta) \quad (14)$$

Stable Layer Formulations ($0 \leq Ri \leq Ri_c^{\text{MOST}}$)

- **Quadratic Form ($Pr = 1$, $\beta_m = \beta_h = \beta$):** Direct substitution of $\zeta = Ri/(1 - \beta Ri)$ into $\phi_m = 1 + \beta\zeta$ yields $\phi_m = (1 - Ri/Ri_c^{\text{MOST}})^{-1}$. Thus, the quadratic Ri -dependence is derived directly from MOST:

$$f_m(Ri) = f_h(Ri) = \begin{cases} \left(1 - \frac{Ri}{Ri_c^{\text{MOST}}}\right)^2, & 0 \leq Ri \leq Ri_c^{\text{MOST}} \\ 0, & Ri > Ri_c^{\text{MOST}} \end{cases} \quad (15)$$

- **Blackadar Linear Form (For Comparison):**

$$f_m(Ri) = f_h(Ri) = \begin{cases} 1 - \frac{Ri}{Ri_c^{\text{MOST}}}, & 0 \leq Ri \leq Ri_c^{\text{MOST}} \\ 0, & Ri > Ri_c^{\text{MOST}} \end{cases} \quad (16)$$